

For each problem round your answer to the nearest hundredth.

1. What is the percent error in using 2.71 as an approximation of e , 2.718?

Using 2.718 for e : $\left| \frac{2.71 - 2.718}{2.718} \right| \times 100 \approx 0.29\%$

2. A dental intervention is estimated to cost \$210. The actual cost is \$320. What is the percentage of error?

$$\left| \frac{210 - 320}{320} \right| \times 100 \approx 34.38\%$$

3. Harry expected to make \$740 in one week after selling some products. He only earned \$560. What is the percentage of error?

$$\left| \frac{740 - 560}{560} \right| \times 100 \approx 32.14\%$$

4. Jim wants to read 30 pages of a book per day. After two weeks, he has only read 22 pages a day. What is the percentage of error?

$$\left| \frac{30 - 22}{22} \right| \times 100 \approx 36.36\%$$

5. You had expected to get \$85 for your birthday to buy a new game, but you only received \$76. What is the percentage of error?

$$\left| \frac{85 - 76}{76} \right| \times 100 \approx 11.84\%$$

6. A company is looking to hire a new manager. The company expected a large number of people to apply for the position. By the end of the hiring process 24 people applied, which was 60% of the expected number of applicants. How many people did the company expect to apply for the job?

$$\frac{60}{100} = \frac{24}{x} \quad x = 40 \text{ applicants}$$

7. What is the percent error in using $\frac{19}{7}$ as an approximation of e , 2.718?

$$\text{Using } e = 2.718 \text{ and } \frac{19}{7} = 2.714: \left| \frac{2.714 - 2.718}{2.718} \right| \times 100 \approx 0.15\%$$

8. There are 2,560 people expected at a rock concert, but only 1,280 turned up. What is the percentage of error?

$$\left| \frac{2,560 - 1,280}{1,280} \right| \times 100 = 100\%$$

9. Justin is preparing for the Gate River Run. His training plan for the first week is shown in the table.

Day of week	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Number of miles (mi.)	4.5	3.0	0	2.5	4.0	0	4.0

By the end of the week, Justin realized he managed to run 12.5% less than he planned. How many miles did Justin run for his first week of training?

$$4.5 + 3.0 + 2.5 + 4.0 + 4.0 = 18 \text{ mi.}$$

$$\frac{100 - 12.5}{100} = \frac{x}{18} \quad x = 15.75 \text{ mi.}$$

10. The population of a town in the year 2010 was 42,000. It was expected to increase its population to 58,000 by the end of the year 2020. But, by that year the population was 53,000. What was the percentage of error?

$$\left| \frac{58,000 - 53,000}{53,000} \right| \times 100 \approx 9.43\%$$